Quiz 22 :-

C22 :-

(A) $\sum_{n=1}^{\infty} \frac{n}{\sqrt[5]{n^7}} = \sum_{n=1}^{\infty} \frac{n}{n^{7/5}}$

$= \sum_{n=1}^{\infty} \frac{1}{n^{7/5-1}} = \sum_{n=1}^{\infty} \frac{1}{n^{2/5}}$ $\quad \frac{7}{5} - 1 = \frac{2}{5}$

P-Series $\quad P = \frac{2}{5} < 1$

$\therefore$ Diverges by P-Series.

(B) $\sum_{n=5}^{\infty} \frac{n+9}{3n+2}$

Divergence test :-

$\lim_{n \to \infty} \frac{n+9}{3n+2} = \frac{1}{3} \neq 0$

$\therefore$ Diverges by Divergence test.

(C) $\sum_{n=1}^{\infty} \frac{e^{2n}}{10^n} = \sum_{n=1}^{\infty} \left(\frac{e^2}{10}\right)^n$

$\therefore \quad |r| = \frac{e^2}{10} < 1$

∴ Converges by Geometric.

(5) $\sum_{n=1}^{\infty} \frac{n^e}{n^\pi}$

C23: $\sum_{n=2}^{\infty} ne^{-n} = \sum_{n=2}^{\infty} \frac{n}{e^n}$

Divergence test :-

$\lim_{n\to\infty} \frac{n}{e^n} = 0$ We know nothing !

Integral test :-

(1) is a_n positive on $[2,\infty)$?
(2) is a_n continuous on $[2,\infty)$?
(3) is a_n decreasing on $[2,\infty)$?

(i) $a_n = \frac{n}{e^n} = \frac{\text{Positive}}{\text{positive}} = \text{positive}$

(ii) $f(x) = \frac{n}{e^n}$, $f'(x) = \frac{e^n - ne^n}{(e^n)^2}$

$\Rightarrow \frac{e^n(1-n)}{(e^n)^2} = \left(\frac{1-n}{e^n}\right) = 0$

$\boxed{n = 1}$

∴ it is continuous on $[2, \infty)$.

(iii) $\lim_{n \to \infty} \left(\frac{1-n}{e^n}\right) = \frac{\text{negative}}{\text{positive}} = \text{negative}$

∴ it is Decreasing on interval $[2, \infty)$

We can use Integral test:-

$$\int_2^\infty f(x)\,dx = \int_2^\infty \frac{x}{e^x}\,dx$$

Let $u = x$, $du = dx$ | $dv = e^{-x}\,dx$, $v = -e^{-x}$

$$= -xe^{-x} + \int_2^\infty -e^{-x}\,dx$$

$$= \lim_{b\to\infty} \left[-xe^{-x} + (-e^{-x}) + C\right]_2^b$$

$$= \lim_{b\to\infty} \left[-xe^{-x} - e^{-x}\right]_2^b$$

$$= \lim_{b\to\infty} \left[-be^{-b} - e^{-b} - \left[-2e^{-2} - e^{-2}\right]\right]$$

$$= \infty$$

$\therefore \sum_{n=2}^{\infty} \frac{n}{e^n}$ is Diverges by Integral test

(C23) $\sum_{n=1}^{\infty} \frac{n^2+1}{n^4+n+3} \Rightarrow$

Comparison test :-

$$\Rightarrow \sum_{n=1}^{\infty} \frac{n^2}{n^4} = \sum_{n=1}^{\infty} \frac{1}{n^2}$$

P-Series

$P = 2 > 1$

Converges

$$\frac{n^2+1}{n^4+n+3} < \frac{n^2}{n^4}$$

$\therefore \sum_{n=1}^{\infty} \frac{n^2+1}{n^4+n+3}$ Converges by Comparison test.

(b) $\sum_{n=5}^{\infty} \frac{4}{n(\ln(n))^3}$

Comparison test :-

$$\Rightarrow \sum_{n=5}^{\infty} \frac{1}{n(\sqrt{n})^3} = \sum_{n=5}^{\infty} \frac{1}{n^{5/2}}$$

P-Series $P = 5/2 > 1$

Converges ∴

$$\sum_{n=5}^{\infty} \frac{4}{n(\ln(n))^3} < \sum_{n=5}^{\infty} \frac{1}{n(\sqrt{n})^3}$$

∴ a_n Converges by Comparison test

(c) $\sum_{n=1}^{\infty} \tan(1/n)$

Comparison test :-

$$\Rightarrow \sum_{n=1}^{\infty}$$